



TheNightOps

Autonomous SRE Agent for Kubernetes

Built with Google Agent Development Kit (ADK) & Gemini



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About Me



Mehul Patel

DevOps Team Lead at BuildingMinds

Google Developer Expert in Google Cloud

Docker Captain

Running Grafana & Friends Meetup in Berlin

Passionate about SRE, Kubernetes, AI-powered operations & Open Source

Agenda



The Problem

Why traditional alerting falls short



What is ADK?

Google Agent Development Kit overview



Architecture

Multi-agent design & MCP integration



Deep Dive

Agents, MCP servers, orchestration patterns



Live Demo

Triggering failures & watching agents investigate



Takeaways

What we learned building with ADK



Me at 3 AM:

"I should automate this"

Also me at 3 AM:

manually SSHing into 14 pods



The Problem

Why traditional incident response doesn't scale

The 3 AM Problem

70%

Alert fatigue rate
in SRE teams

45min

Average time to first
meaningful action

60%

Incidents repeat with
same root cause

Alert → Triage → Investigate → Escalate → Fix → Repeat

What if an AI agent could handle the investigation autonomously?



SRE Job Description:

"Automate everything"

SRE Reality:

"kubectl get pods -A | grep -v Running"

...at 3 AM, for the 47th time this quarter



What is Google ADK?

Agent Development Kit — building blocks for multi-agent systems

Google ADK at a Glance

Open-source framework for building multi-agent AI systems

Open Source

Apache 2.0

Python, Go, TS & Java

SDK support

Model-Agnostic

Gemini, Claude, GPT-4o, Mistral

Production-Ready

Built-in eval & debugging



Multi-Agent Orchestration

Parallel, sequential & hierarchical agent orchestration with structured handoffs



Native MCP Support

Connect official & custom MCP servers for tool integration



Streaming & Real-time

Bidirectional audio/video streaming with real-time agent interaction



Local Dev & Deploy

CLI + Web UI for development, deploy to Cloud Run or GKE

Deep GCP Integration

ADK fits naturally into the Google Cloud ecosystem

Compute

**Cloud Run, GKE
Compute Engine**

Serverless & container
deployment

Data

**BigQuery, AlloyDB
Cloud Spanner, Postgres**

Storage & analytics
integration

Operations

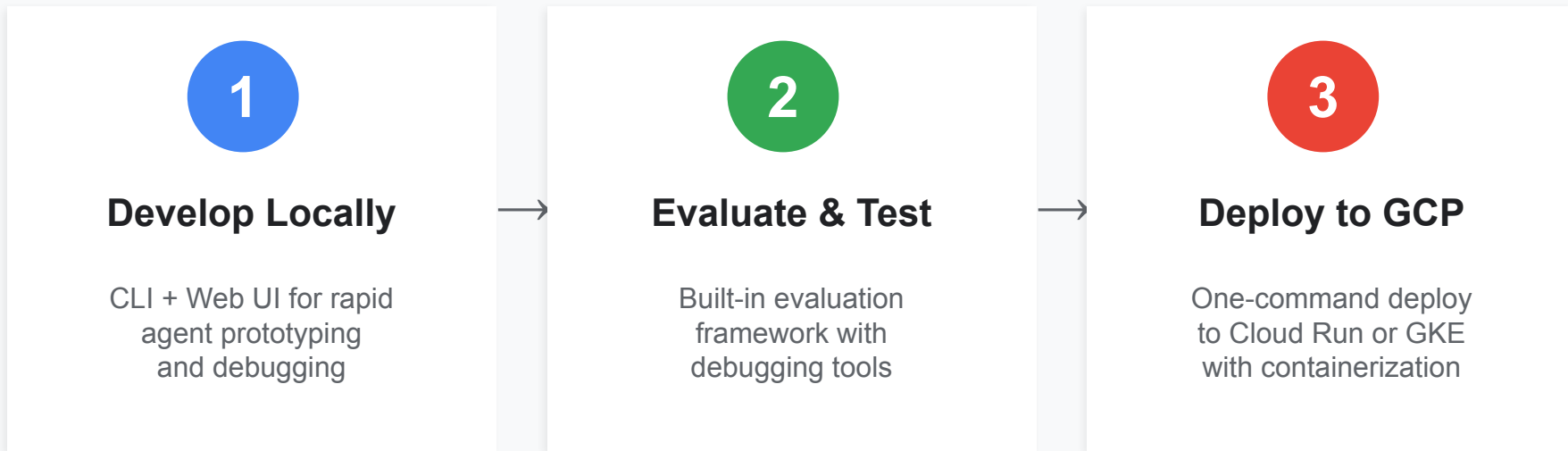
**Cloud Monitoring
Cloud Logging, Security**

Observability &
compliance built-in

+ Apigee for API management | Vertex AI for model serving | Cloud IAM for security

ADK Development Workflow

From local development to production in three steps



Why ADK Matters:

Unified enterprise AI framework

All GCP services no duplication

Enterprise app connectivity

Production-ready from day one

Why ADK for SRE?

Traditional Alerting

- Static threshold rules
- Manual investigation per alert
- Context switching between tools
- No memory of past incidents
- Linear, single-threaded process

ADK-Powered Agent

- AI-driven pattern recognition
- Parallel multi-agent investigation
- Unified tool access via MCP servers
- TF-IDF incident memory & matching
- Orchestrated, concurrent workflows

Incident Response Expectation

1. Get alert
2. Check dashboard
3. Find root cause
4. Fix it
5. Go back to sleep

Incident Response Reality

1. Get alert
2. Which cluster was this again?
3. VPN dropped
4. Wrong terminal, that was prod
5. Slack: "is anyone else seeing this?"
6. It was DNS. It's always DNS.
7. Sun is rising



Architecture

Multi-agent design with ADK orchestration & MCP tools

Plan A — The Dream

Original Vision

Full multi-agent MCP architecture — 1 orchestrator + 5 sub-agents + 5 MCP servers



ADK Root Orchestrator + 5 Sub-Agents

Powered by Gemini

↓ All tools via MCP ↓

GC Observability MCP

Cloud Logging
Monitoring, Trace

official

Google GKE MCP

Pod status
events, deploys

official

Grafana MCP

Alerts, dashboards
annotations

official

Slack MCP

Channel alerts
thread updates

custom

Notifications MCP

Email, Telegram
WhatsApp

custom



Reality Check:

GKE & Observability MCPs are in preview/beta. In testing:

MCP connections hanging/timing out • ADK TaskGroup/ExceptionGroup errors on MCP failures
Investigations stuck at "Triage" phase • httpx.AsyncClient connection pool issues

Plan B — Simple Mode

Working Today

1 agent + 10 kubectl FunctionTools — no MCP, no sub-agents, just kubectl

INGESTION

Grafana / Alertmanager / PagerDuty → Webhook Receiver → Dedup



ADK AGENT

Single agent — no orchestrator, no sub-agents

Single Agent (Gemini 3.1 Pro Preview)

▼ kubectl subprocess calls ▼

10 kubectl FunctionTools

Read-only RCA — no write operations

get_pods

get_events

get_deployments

top_pods

get_pod_logs

describe_pod

get_deployment_history

get_namespaces

get_nodes

get_resource_yaml



Output: Root Cause Analysis (read-only)

Root Orchestrator

The brain of the operation — powered by Gemini

**Alert
Received**

**Root
Orchestrator**

**Parallel
Sub-agents**

**Synthesize
& Act**

How it works:

- Receives deduplicated alert with fingerprint and context
- Delegates to Log Analyst, Deployment Correlator, Runbook Retriever in parallel
- Synthesizes findings using Gemini's reasoning to generate an RCA
- Proposes remediation actions, routed through the Policy Engine
- Drafts and sends notifications via Communication Drafter

Specialized Sub-Agents



Log Analyst

Queries GC Observability MCP for error patterns, stack traces, and anomalies



Deployment Correlator

Uses GKE MCP to check pod health, events, deployments, and what changed



Runbook Retriever

Searches Grafana MCP for alerts, dashboards, and past incident runbooks



Communication Drafter

Generates RCAs and sends via Slack MCP + Notifications MCP

MCP Servers: The Tool Layer

Model Context Protocol — how agents talk to infrastructure

Official MCP Servers



Google Cloud Observability MCP

Cloud Logging, Monitoring,
Trace — remote, IAM auth



Google GKE MCP

Pod status, events,
deployments — remote



Grafana MCP

Alerts, dashboards,
annotations — stdio, token

Custom MCP Servers



Slack MCP

Post incident alerts, thread updates,
channel notifications — custom built



Notifications MCP

Email (SMTP), Telegram, WhatsApp
multi-channel delivery — custom built

"How many agents does your system have?"

Yes.

Log Analyst

Deployment Correlator

Runbook Retriever

Communication Drafter

Root Orchestrator

Anomaly Detector

Intelligence Layer



Incident Memory

- TF-IDF vectorization of past incidents
- Cosine similarity matching
- Learns from every investigation
- Surfaces relevant runbooks automatically



Policy Engine

4 approval levels:

- Auto-approve — safe actions (restart pod)
- Env-gated — auto in dev, manual in prod
- Always-approve — human required
- Blocked — never auto-execute

Ingestion Pipeline

Grafana

Alertmanager

PagerDuty

Generic



Webhook Receiver

FastAPI on :8090
normalizes payload format

Fingerprint & Dedup

`compute_fingerprint()`
sliding window dedup

K8s Event Watcher

Continuous stream of
cluster events as context



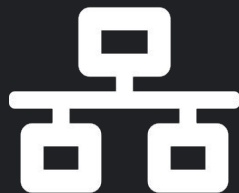
When official MCP servers exist for GKE and Observability:

"npm install and done!"

When you need Slack + Notifications MCP:

"It ain't much, but it's honest work"

(400 lines of Python later...)



Before MCP:

"Let me write 47 custom API integrations"

After MCP:

"npm install @google-cloud/gcloud-mcp"

Official Google Cloud + Grafana MCPs = plug and play



A moment of silence...

for the Demo Gods

"Please work. Please work. Please work."

It worked on my machine. Yesterday. Once.



Live Demo

Let's break things and watch the agents investigate

Demo Scenarios

1

memory-leak

Gradual memory increase until OOM

2

cpu-spike

Sudden CPU utilization spike

3

cascading-failure

Multi-service dependency chain failure

4

config-drift

Configuration mismatch detection

5

oom-kill

Container killed by kernel OOM

Demo Commands

Quick reference for the live demo

Start agent

```
$ nightops agent watch
```

Interactive

```
$ nightops agent run --interactive
```

Trigger

```
$ ./scripts/demo-gke.sh --scenario cpu-spike
```

Dashboard

```
$ nightops dashboard
```

Verify

```
$ nightops verify
```

What to Watch For

During the live demo, pay attention to these key moments

- | | | |
|---|----------------------------------|---|
| 1 | Alert arrives via webhook | Watch the dedup engine filter noise |
| 2 | Orchestrator delegates | Sub-agents spin up in parallel |
| 3 | MCP servers query infra | GKE, Observability & Grafana MCPs fetch live data |
| 4 | RCA generation | Gemini synthesizes findings into root cause |
| 5 | Notification sent | Slack MCP + Notifications MCP deliver the report |



Production:

"This is fine."

TheNightOps:

"Already identified root cause, drafted RCA,
and notified the team. Go back to sleep."



MTTR with manual investigation:

45 minutes

MTTR with TheNightOps:

< 3 minutes

...and your SRE actually got to sleep

Key Takeaways



ADK makes multi-agent systems accessible

Clean abstractions for orchestration, sub-agents, and tool integration



MCP is the universal tool protocol

Official Google Cloud & Grafana MCPs + custom servers = plug and play



Gemini excels at reasoning over SRE data

Pattern recognition across logs, events, and deployment history



Start with guardrails, not YOLO

Policy Engine ensures safe, graduated automation from day one



The real takeaway from this talk:

Let the agents handle 3 AM.

You deserve sleep.

(Your on-call rotation will thank you)

Get Started

1

Clone & Install

```
git clone github.com/nomadicmehul/TheNightOps
```

2

Configure

```
cp config/.env.example config/.env
```

3

Verify

```
nightops verify
```

4

Run Demo

```
./scripts/demo-gke.sh --scenario memory-leak
```

5

Explore

```
nightops agent run --interactive
```



Thank You!

Questions & Discussion

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